

Marcar como hecha

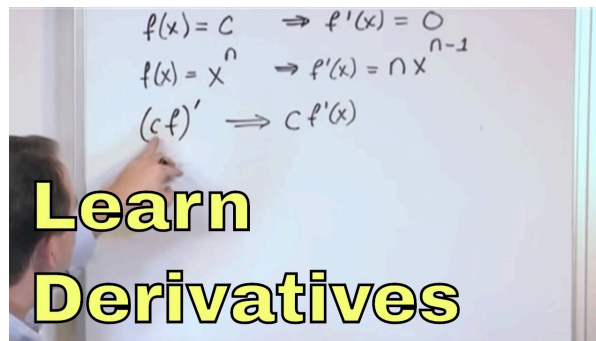
Resumir

If you are interested in stepping into Artificial Intelligence, Machine Learning, and Data Science, start with the materials recommended by our mentors.

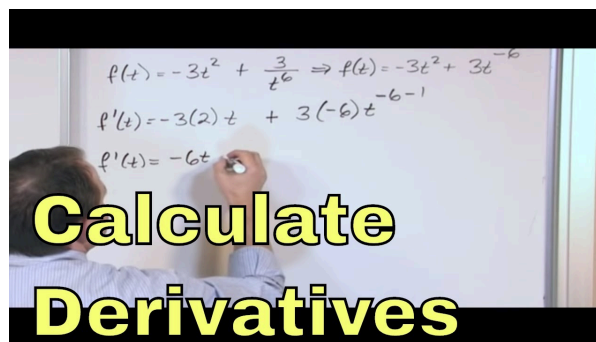
## Fundamentals of Mathematics

### Derivatives

- Learn

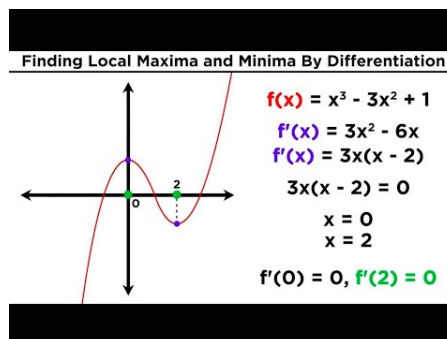


- Learn to



### Maxima and Minima

- 



- [Learn how to find Maxima and Minima using Derivatives](#)

### Differential equations

- [Differential equations introduction](#)

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## LINEAR ALGEBRA

- Comprehensive beginners [guide to Linear Algebra for Data Scientists](#)
- A comprehensive [course on Linear Algebra](#) covering such topics as vectors and matrices, eigenvalues and eigenvectors, and their implementation in working with datasets. The course aims to help students bridge the gap into linear algebra problems, and understand how to apply these concepts to machine learning.

## Probability theory fundamentals

### Probability theory

- A visualized intro to the [basic concepts of probability theory](#)



**Basics of Probability: Unions, Inters**  
jbststatistics



Ver en

- in seven bite-sized videos



by Crash Course with real-life examples



**Lecture 1: Probability and Counting**  
Harvard University

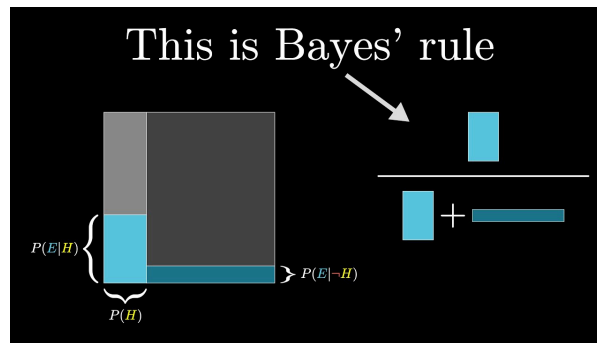


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- by Harvard educators

### Bayes inference

- An [introduction to Bayesian methods + probabilistic programming](#) with a computation/understanding-first, mathematics-second point of view. All in pure Python.
- An explanation of the



, which is believed to be the most important one in probability theory

- An introduction to [Bayes Theorem for Machine Learning](#)

## Statistics

### Basic Concepts

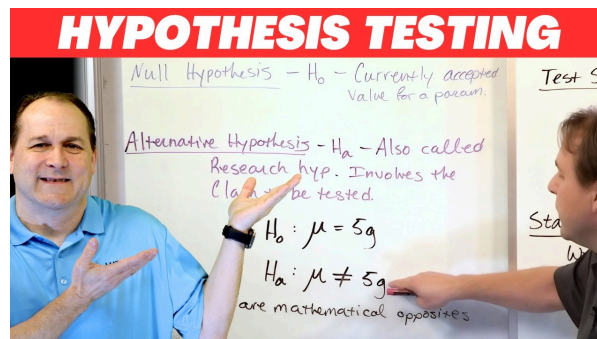
- [Introduction to Statistics for Data Science](#)
- Eight-hour course that covers the



, and introduces the various methods used to collect, organize, summarize, interpret and reach conclusions about data

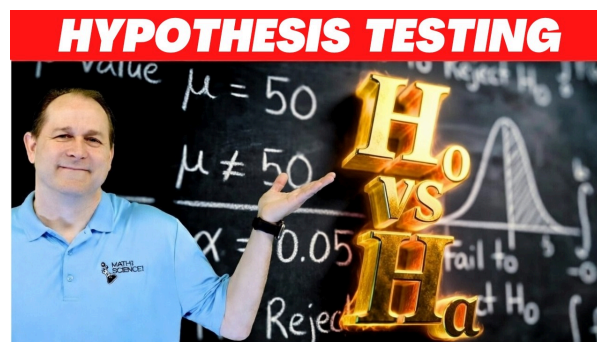
### Hypothesis Testing

- An Introduction to [Statistical Hypothesis Testing](#)
- Step-by-step



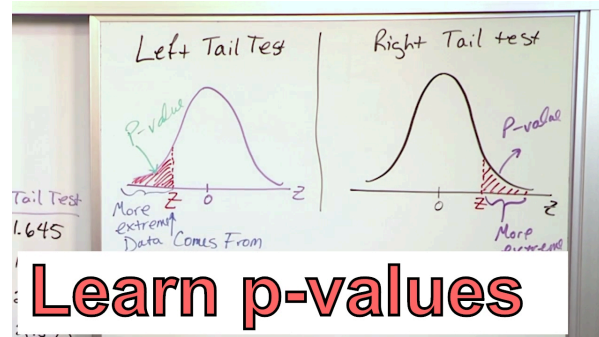
that teaches students how to perform hypothesis testing in statistics by working examples and solved problems

- An explanation of



as part of a hypothesis test in statistics

- A video tutorial that explains



## Maximum Likelihood Estimation

- A brief introduction to [Maximum Likelihood Estimation for Machine Learning](#)

# OPTIMIZATION THEORY

## Optimization for AI, ML, and DS

Optimization is one of the three pillars that AI, ML, DS professionals must understand thoroughly. Familiarize yourself with [its fundamentals](#).

## Algorithms and data structures

### Data structures

- [Data Structures & Algorithm Basic Concepts](#)

### Sorting algorithms

- Introduction to [sorting techniques in Data Structure](#)

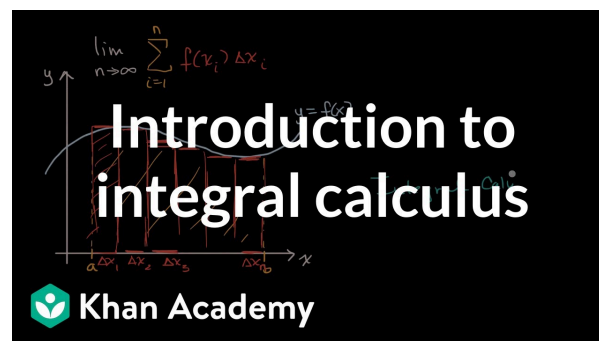
## Basics of Python

- A quick [crash course](#) on both the Python programming language and its use for scientific computing

## ADDITIONAL TOPICS

### INTEGRALS

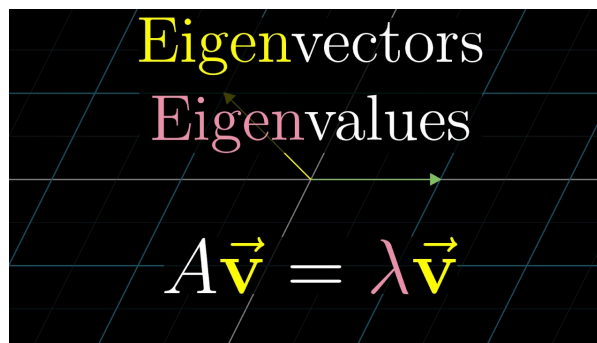
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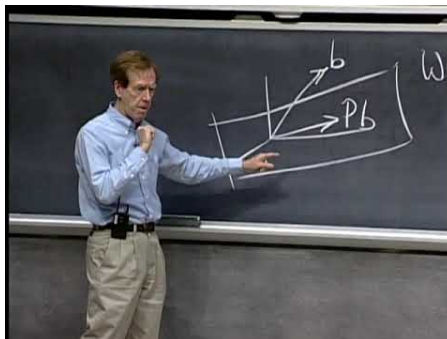
- [Integrals comprehensive tutorial](#)

## EIGENVECTORS AND EIGENVALUES

- An easy-to-digest

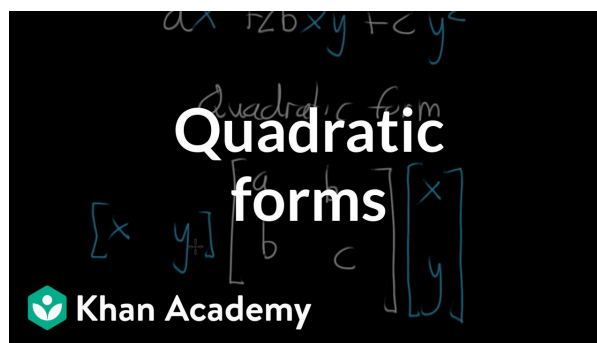


- Advanced level



## QUADRATIC FORMS

- An explanation of



Última modificación: viernes, 20 de octubre de 2023, 13:18